HWRS 505: Vadose Zone Hydrology

Steady-state 1D unsat flow (Part 1)

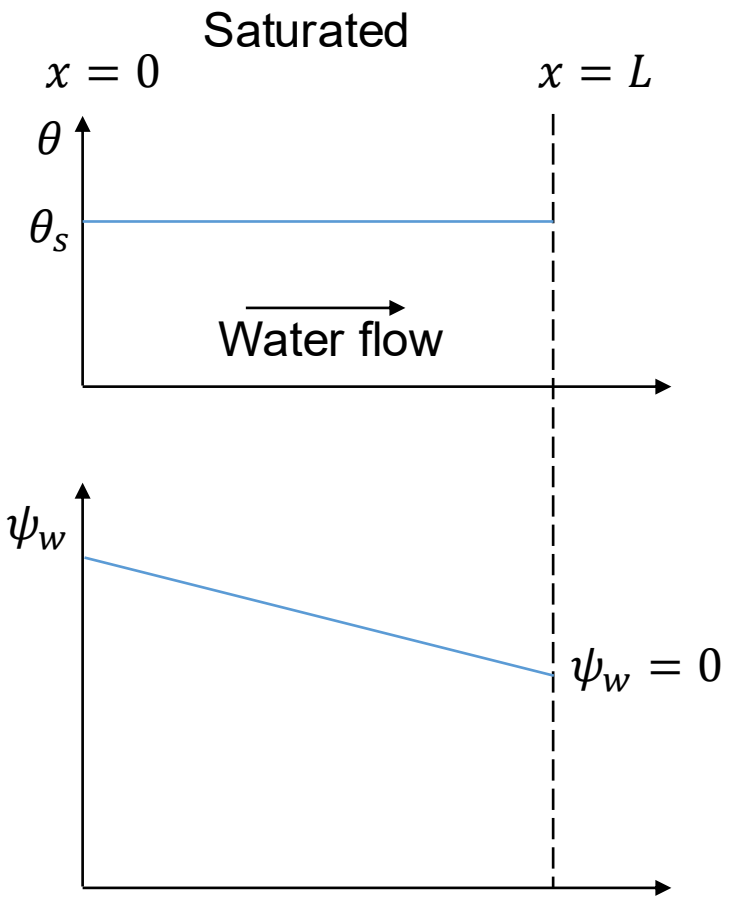
Agenda: Steady-state unsaturated flow
Reading: Chapter 11 (Pinder & Celia, 2006)

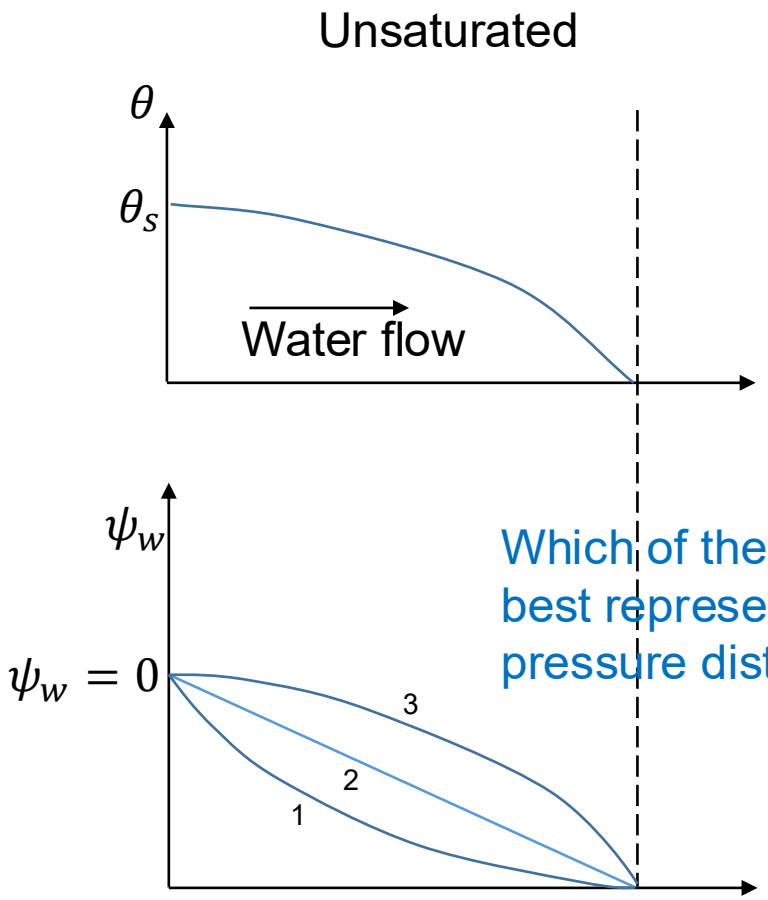
Review

- Three forms of Richards' equation
  - Mixed form
  - Pressure head-based form
    - ✓ Specific moisture capacity
  - Water content-based form
    - ✓ Soil moisture diffusivity (What does the equation have to do with "diffusion"?)
    - ✓ Cannot be used if the domain involves saturated water flow
  - How to include soil and fluid compressibility?
- Richards' assumptions
  - Air pressure remains almost zero everywhere, but air does move.
  - Does "air movement" make the Richards' equation invalid? No, as long as air pressure remains almost zero everywhere.

Steady-State Unsaturated Flow

Horizontal flow (Homogeneous column)





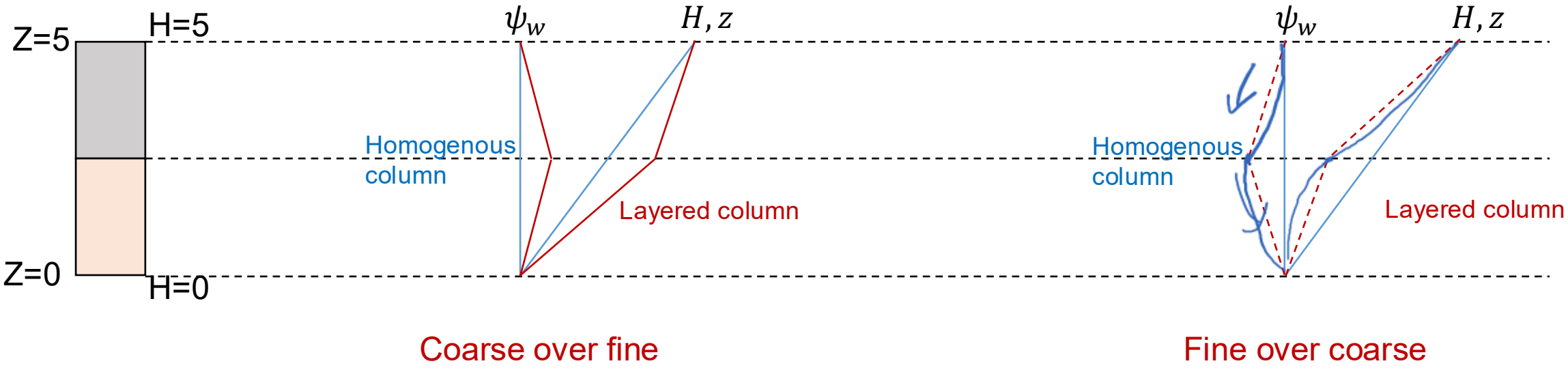
Which of the curves (1, 2, or 3) best represents the water pressure distribution?

Answer: 3

- Unsaturated flow involves nonlinearities that make their behaviors differ from that of the saturated flow

Steady-State Unsaturated Flow

Vertical flow (in layered columns)



Coarse over fine

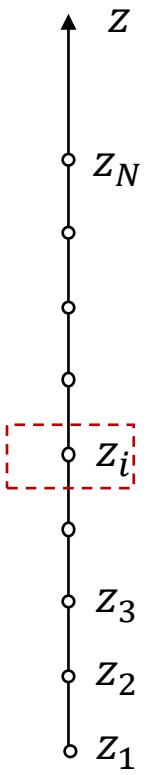
Fine over coarse

- Hydraulic head and pressure head are both continuous in space.
- Is water saturation continuous?

Steady-State Unsaturated Flow: Numerical Soln.

$$\cancelto{0}{\frac{\partial \theta_w}{\partial t}} - \frac{\partial}{\partial z}\left(K\frac{\partial \psi_w}{\partial z}\right) - \frac{\partial K}{\partial z} = 0 \quad \xrightarrow{\text{Steady-state}} \quad \underbrace{\frac{\mathrm{d}}{\mathrm{d}z}\left(K\frac{\mathrm{d}\psi_w}{\mathrm{d}z}\right)}_{1} + \underbrace{\frac{\mathrm{d}K}{\mathrm{d}z}}_{2} = 0 \qquad (1)$$

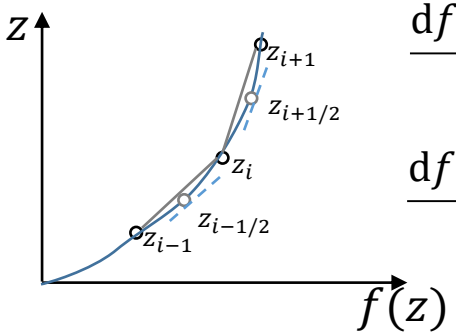
- Equation (1) is a second-order ordinary differential equation in 1D.
- It is nonlinear because $K = K(\psi_w)$ is a nonlinear function.
- To solve it, we need <u>two boundary conditions</u> and we need to do it in an <u>iterative</u> procedure.



- How to solve this 1D **nonlinear ordinary differential equation**?

- <u>Key idea</u>: Divide the domain into many boxes and convert the **differential equation** to a system of nonlinear **algebraic equations**.

- Technique: Use finite difference to approximate derivatives

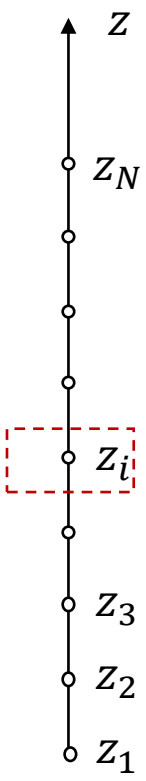


$$\frac{\mathrm{d}f(z_{i+1/2})}{\mathrm{d}z} \approx \frac{f(z_{i+1}) - f(z_i)}{z_{i+1} - z_i}$$

$$\frac{\mathrm{d}f(z_{i-1/2})}{\mathrm{d}z} \approx \frac{f(z_i) - f(z_{i-1})}{z_i - z_{i-1}}$$

Steady-State Unsaturated Flow: Numerical Soln.

$$\cancel{\frac{\partial \theta_w}{\partial t}}^{\,0} - \frac{\partial}{\partial z}\left(K\frac{\partial \psi_w}{\partial z}\right) - \frac{\partial K}{\partial z} = 0 \quad \xrightarrow{\text{Steady-state}} \quad \underbrace{\frac{\mathrm{d}}{\mathrm{d}z}\left(K\frac{\mathrm{d}\psi_w}{\mathrm{d}z}\right)}_{1} + \underbrace{\frac{\mathrm{d}K}{\mathrm{d}z}}_{2} = 0 \qquad (1)$$



$$\underbrace{\frac{\mathrm{d}}{\mathrm{d}z}\left(K\frac{\mathrm{d}\psi_w}{\mathrm{d}z}\right)\Big|_{z_i}}_{1} \approx \frac{\left(K\frac{\mathrm{d}\psi_w}{\mathrm{d}z}\right)_{i+1/2} - \left(K\frac{\mathrm{d}\psi_w}{\mathrm{d}z}\right)_{i-1/2}}{\Delta z} \approx \frac{K_{i+1/2}\frac{\psi_{w,i+1}-\psi_{w,i}}{\Delta z} - K_{i-1/2}\frac{\psi_{w,i}-\psi_{w,i-1}}{\Delta z}}{\Delta z}$$

$$= \frac{K_{i+1/2}(\psi_{w,i+1}-\psi_{w,i}) - K_{i-1/2}(\psi_{w,i}-\psi_{w,i-1})}{\Delta z^2}$$

$$\underbrace{\frac{\mathrm{d}K}{\mathrm{d}z}\Big|_{z_i}}_{2} \approx \frac{K_{i+1/2}-K_{i-1/2}}{\Delta z}$$

$$\Rightarrow \frac{K_{i+1/2}}{\Delta z^2}(\psi_{w,i+1}-\psi_{w,i}) - \frac{K_{i-1/2}}{\Delta z^2}(\psi_{w,i}-\psi_{w,i-1}) + K_{i+1/2}\frac{1}{\Delta z} - K_{i-1/2}\frac{1}{\Delta z} = 0$$

$$\Rightarrow \frac{K_{i+1/2}}{\Delta z^2}\psi_{w,i+1} - \left(\frac{K_{i+1/2}}{\Delta z^2} + \frac{K_{i-1/2}}{\Delta z^2}\right)\psi_{w,i} + \frac{K_{i-1/2}}{\Delta z^2}\psi_{w,i-1} + K_{i+1/2}\frac{1}{\Delta z} - K_{i-1/2}\frac{1}{\Delta z} = 0$$

This is an algebraic equation with 3 unknowns. We can write such an algebraic equation for each node or box and we can get N algebraic equations for the N unknowns $(\psi_{w,1}, \psi_{w,2}, \ldots, \psi_{w,N})$

Steady-State Unsaturated Flow: Numerical Soln.

In matrix form:

$$
\begin{pmatrix}
1 & 0 & 0 & & & & & \\
\frac{K_{3/2}}{\Delta z^2} & -\frac{(K_{3/2}+K_{5/2})}{\Delta z^2} & \frac{K_{5/2}}{\Delta z^2} & & & & & \\
 & \frac{K_{5/2}}{\Delta z^2} & -\frac{(K_{5/2}+K_{7/2})}{\Delta z^2} & \frac{K_{7/2}}{\Delta z^2} & & & & \\
 & & \ddots & \ddots & \ddots & & & \\
 & & & \frac{K_{i-1/2}}{\Delta z^2} & -\frac{(K_{i-1/2}+K_{i+1/2})}{\Delta z^2} & \frac{K_{i+1/2}}{\Delta z^2} & & \\
 & & & & \ddots & \ddots & \ddots & \\
 & & & & \frac{K_{n-1-1/2}}{\Delta z^2} & -\frac{(K_{n-1-1/2}+K_{n-1+1/2})}{\Delta z^2} & \frac{K_{n-1+1/2}}{\Delta z^2} \\
 & & & & & 0 & 0 & 1
\end{pmatrix}
\begin{pmatrix}
\psi_{w,1} \\ \psi_{w,2} \\ \psi_{w,3} \\ \vdots \\ \psi_{w,i} \\ \vdots \\ \psi_{w,n-1} \\ \psi_{w,n}
\end{pmatrix}
\begin{pmatrix}
B.C. \\ \frac{K_{3/2}-K_{5/2}}{\Delta z} \\ \frac{K_{5/2}-K_{7/2}}{\Delta z} \\ \vdots \\ \frac{K_{i-1/2}-K_{i+1/2}}{\Delta z} \\ \vdots \\ \frac{K_{n-1-1/2}-K_{n-1+1/2}}{\Delta z} \\ B.C.
\end{pmatrix}
$$

$\Rightarrow \boldsymbol{F}(\boldsymbol{\psi}_w) \cdot \boldsymbol{\psi}_w = \boldsymbol{R}$ How to solve this nonlinear system of algebraic equations?

=> Employ an iterative methods